Data Preprocessing Report

Online Retail II (UCI dataset 502) · Business Analytics, M.Sc Semester 3 · techniques applied from *BA-6 Preprocessing Techniques*

1,067,371	1,028,743	38,628	96.4%	16
ROWS IN	ROWS OUT	ROWS REMOVED	RETAINED	TECHNIQUES

1. What this document is

The dashboard shows what the data says. This document shows why the data can be trusted. It records every transformation applied to the raw workbook, the technique from the course PDF that each one implements, the number of rows it affected, and the reasoning behind the choice - including the cases where the obvious action would have been the wrong one.

The raw file holds **1,067,371 rows** across two worksheets covering December 2009 to December 2011. After preprocessing, **1,028,743 rows** remain (96.38%), carrying **GBP 18,854,778.33** of net revenue against **GBP 19,287,250.57** in the raw file. The GBP 432,472.24 difference is not lost data - it is revenue that was being counted twice.

2. Three problems that had to be found first

> **The two worksheets overlap, and a naive append double-counts**
>
> The sheet named `Year 2009-2010` runs to 9 December 2010, but `Year 2010-2011` starts on 1 December 2010. **1,088 invoices - 22,202 rows - appear in both.** They are not similar records; they are identical on invoice number, stock code, quantity, timestamp, price and customer. Verified line by line, not inferred from the date range.
>
> Anyone who stacks these two sheets without checking reports **GBP 376,266.82** of revenue that does not exist. This is the single most consequential finding in the whole exercise, and nothing in the file announces it.

> **The missing descriptions are provably Missing at Random**
>
> 4,382 rows have no `Description`. Of those, **100% also have no Customer ID and 100% have a price of zero**. Missingness is completely predicted by other observed columns, which is the textbook definition of MAR rather than MCAR. That is a demonstrated classification, not a judgement call - and because the rows carry no revenue, deleting them costs nothing.

> **One product can have several names**
>
> **1,210 of 5,132 stock codes carry more than one description.** Code 23298 is both `BUNTING , SPOTTY` and `SPOTTY BUNTING` ; 23203 is both `JUMBO BAG DOILEY PATTERNS` and `JUMBO BAG VINTAGE DOILY` . Two consequences: an exact eight-column duplicate check misses lines repeated under a second spelling, and grouping revenue by description instead of by stock code silently splits one product across several rows. The product dimension is therefore keyed on `StockCode` , never on the name.

3. Every step, in order

Row counts are stated only for steps that actually delete rows. Derivation, schema and aggregation steps leave the row count untouched; reporting them as deletions would suggest a million rows were discarded in order to build a 604-row daily summary.

#	Technique	PDF section	Type	Before	After	Removed
0	Raw load (both sheets unioned)	Data Integration	Row filter	1,067,371	1,067,371	–
1	Cross-sheet overlap resolution	Data Integration	Row filter	1,067,371	1,045,169	22,202
2	Duplicate removal (within sheet)	Duplicate Removal	Row filter	1,045,169	1,033,036	12,133
3	Unit conversion & format standardisation	Unit Conversion	Derivation	–	–	–
4	Duplicate removal (transaction business key)	Duplicate Removal	Row filter	1,033,036	1,033,034	2
5	Missing-value removal (MAR, zero-value)	Missing Value Types (MAR)	Row filter	1,033,034	1,028,759	4,275
6	Missing Description imputation (placeholder)	Missing Value Handling	Derivation	–	–	–
7	Test-record removal	Data Cleaning (anomalies)	Row filter	1,028,759	1,028,743	16
8	Attribute construction	Attribute Construction	Derivation	–	–	–
9	Outlier detection (z-score + IQR)	Z-score Normalization	Derivation	–	–	–
10	Listwise deletion (customer table only)	Listwise Deletion	Parallel table	–	–	–
11	Normalization (Min-Max + Z-score)	Data Normalization	Derivation	–	–	–
12	CRM segmentation & CLV	CRM Strategies (BA-2)	Derivation	–	–	–
13	Discretization & generalization (RFM)	Discretization / Generalization	Derivation	–	–	–
14	Dimensionality reduction (star schema)	Dimensionality Reduction	Schema	–	–	–
15	Noise reduction (MA7 + exponential smoothing)	Noise Reduction	Derivation	–	–	–

#	Technique	PDF section	Type	Before	After	Removed
16	Data cube aggregation	Data Cube Aggregation	Aggregation	–	–	–
17	Histogram reduction	Histogram Reduction	Aggregation	–	–	–

4. Why each step was done that way

0. Raw load (both sheets unioned)

PDF: Data Integration
Year 2009-2010 and Year 2010-2011 stacked into one table.

1. Cross-sheet overlap resolution

PDF: Data Integration
Sheets overlap 1-9 Dec 2010; the same invoices appear twice. Removing the second copy corrects GBP 376,266.82 of double-counted revenue.

2. Duplicate removal (within sheet)

PDF: Duplicate Removal
Exact repeats of all 8 fields inside one sheet. Removed GBP 55,450.05 of double-counted revenue.

3. Unit conversion & format standardisation

PDF: Unit Conversion

19,247 rows remapped to canonical country names (EIRE->Ireland, RSA->South Africa, Unspecified->Unknown) + ISO-3 codes

Trimmed and title-cased Description, upper-cased StockCode, mapped 19,247 rows to canonical country names with ISO-3 codes, cast Invoice and Customer ID to text (identifiers, not numbers).

4. Duplicate removal (transaction business key)

PDF: Duplicate Removal
1,188 StockCodes carry more than one Description (e.g. 23298 is both 'Bunting , Spotty' and 'Spotty Bunting'), so an exact 8-column match misses lines duplicated under a second spelling. De-duplicating on Invoice+StockCode+Quantity+Price+Timestamp+Customer caught 2 further rows worth GBP 551.85. This is also why Dim_Product is keyed on StockCode, never on Description.

5. Missing-value removal (MAR, zero-value)

PDF: Missing Value Types (MAR)
Rows with no Description AND Price = 0. Missingness is fully predicted by observed fields, so MAR not MCAR. Revenue impact GBP 0.00 (zero by construction) - deleting them loses nothing.

6. Missing Description imputation (placeholder)

PDF: Missing Value Handling 0 rows imputed

Safeguard for any Description still missing after step 4: fill with 'Unknown Item' rather than delete, since such rows carry real revenue. 0 rows needed it - every missing Description was a zero-value record already removed, which is itself the evidence for the MAR claim.

7. Test-record removal

PDF: Data Cleaning (anomalies)

StockCode TEST001/TEST002 are system tests, not transactions.

8. Attribute construction

PDF: Attribute Construction 14 new attributes added

Derived Revenue = Quantity x Price, plus the IsCancelled, IsReturn, IsProduct and HasCustomer flags, the ItemType and TransactionType labels, and seven calendar attributes. Mirrors the PDF's Defect Percentage = Defective / Produced x 100.

9. Outlier detection (z-score + IQR)

PDF: Z-score Normalization 1,751 rows flagged, 0 deleted

1,751 rows flagged (both |z| > 3 AND beyond 3xIQR), carrying GBP 401,840.79. FLAGGED, NOT DELETED - the 80,995-unit order and the GBP 38,970 Manual adjustment are real events. A plain 1.5xIQR rule would have discarded ~11% of rows.

10. Listwise deletion (customer table only)

PDF: Listwise Deletion fact keeps 1,028,743; customer table built from 797,867

230,876 rows have no Customer ID but carry GBP 2,565,542.41. They stay in the fact table and are dropped only for customer analysis - you cannot impute an identity. This is the dual-grain decision.

11. Normalization (Min-Max + Z-score)

PDF: Data Normalization 5,878 scored customers scaled, 6 new columns

Recency, Frequency and Monetary each scaled to 0-1 (Min-Max) and to standard-deviation units (Z-score) before scoring, so no variable dominates on units alone.

12. CRM segmentation & CLV

PDF: CRM Strategies (BA-2) 8 CRM strategies, 3 churn states, CLV per customer

Each RFM segment mapped to one of the eight CRM strategies in BA-2 CRM.pdf with a recommended action; churn status derived from recency (the PDF's telecom case); CLV computed with the PDF's own formula, annualising purchase frequency over observed tenure and assuming a 3-year lifespan.

13. Discretization & generalization (RFM)

PDF: Discretization / Generalization 9 scored segments (+1 unscored), 5 recency bands, 3 value tiers

Continuous R, F, M cut into quintile scores 1-5, then generalized into 10 named segments, 5 recency bands and 3 value tiers - the PDF's Young / Middle-aged / Senior pattern.

14. Dimensionality reduction (star schema)

PDF: Dimensionality Reduction `29 columns -> 17 fact columns + 6 dimensions`
One flat 29-column table split into a 17-column fact plus 5 dimensions. Descriptive attributes stored once per key instead of once per row. Note Revenue = Quantity x Price is deliberate redundancy - never feed all three into one correlation.

15. Noise reduction (MA7 + exponential smoothing)

PDF: Noise Reduction `2 smoothed series over 604 daily points`
Daily revenue is spiky. Added a 7-point moving average and an exponentially smoothed series with alpha = 0.3, exactly the PDF's New = a*Actual + (1-a)*Previous formula.

16. Data cube aggregation

PDF: Data Cube Aggregation `1,028,743 lines -> 67,787 summary rows (15x smaller)`
Line-level rows pre-summarised into daily (604), month x country (594) and month x product (66,589) cubes so dashboard pages scan thousands of rows, not a million.

17. Histogram reduction

PDF: Histogram Reduction `1,078,080 values -> 23 bin frequencies`
Distributions of Quantity, Unit Price and Basket Value stored as 23 bin frequencies instead of 1,078,080 raw values, preserving shape at a fraction of the size.

5. The three decisions worth defending

Dual grain: 22% of rows kept, and dropped, at the same time

230,876 rows (22.4%) have no Customer ID, and they carry GBP 2,565,542.41 - 13.6% of net revenue. Listwise deletion across the whole dataset, the reflex response, would delete a seventh of the business and every revenue figure would be wrong.

So deletion is applied at one grain only. The fact table keeps every row, so revenue, orders and product analysis are complete. The customer table is built by complete-case analysis, because there is no honest way to impute an identity - a customer ID is not a measurement with a plausible average, it is a name. The dashboard reports the unidentified share openly rather than hiding it.

Outliers are flagged, never deleted

A plain 1.5×IQR rule flags **116,489 rows on quantity alone - about 11% of the data**. A rule that discards a ninth of a dataset is not finding errors; it is describing a naturally skewed retail distribution, where most baskets are small and a few are wholesale.

The test used here is deliberately strict: a row is flagged only if it is **both** beyond three standard deviations **and** outside 3×IQR. That flags **1,751 rows** carrying GBP 401,840.79 (2.1% of revenue). None are deleted. The 80,995-unit order and its matching cancellation are a real transaction and a real reversal; the GBP 38,970 `M` line is a real manual adjustment. Deleting them would be editing the business to suit the statistics.

Zero-price rows are separated, not swept up

6,202 rows have a price of zero. 4,275 of them also lack a description and a customer, and were removed as non-transactions. The remainder are real events - promotions, samples and adjustments - so they are retained and labelled `Zero-price (promo/adjustment)` in `TransactionType`, where they can be included or excluded deliberately rather than by accident.

6. What the cleaned data looks like

RFM segment	Customers	Share
Hibernating / Lost	1,664	28.0%
Champions	1,324	22.3%
Loyal Customers	1,206	20.3%
New / Promising	560	9.4%
At Risk	454	7.6%
Needs Attention	441	7.4%
Others	161	2.7%
Cannot Lose Them	68	1.1%
No Net Value (returns / cancellations only)	62	1.0%
Unknown / Guest (not identified)	1	0.0%

7. Validation

The pipeline is not trusted on the strength of having run without an error. `01_scripts/04_validate.py` asserts 35 conditions and exits non-zero if any fail: no orphaned keys between the fact table and any dimension, no nulls in any join column, `Revenue = Quantity × Price` to the penny on every row, each aggregate cube reconciling to the fact table's total, each audit step's closing row count equal to the next step's opening count, and normalized values landing inside their expected ranges. All 35 pass.

8. Honest limitations

- All amounts are GBP; no exchange rates are applied to overseas orders.
- There is no cost data, so margin cannot be computed - only revenue.
- Two years is too short to separate a genuine trend from a seasonal cycle.
- The data is heavily UK-weighted, so country comparisons are indicative.
- Churn is inferred from recency; the dataset records no cancellations of a relationship, only of orders.
- CLV assumes a three-year customer lifespan, stated rather than measured, because the observation window is shorter than a realistic lifespan.

Generated from `02_clean/Dim_DataQuality.csv` and `02_clean/Fact_Sales.parquet` by `01_scripts/07_build_preprocessing_report.py`. Every figure is read from the data at build time; none is typed by hand.